Yacin Ben Kacem

Final-Year Software Engineering Student — Secure Agentic AI & Retrieval-Augmented Generation (RAG) Systems
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SUMMARY

Final-year Software Engineering student at ENISO specializing in agentic AI and Retrieval-Augmented Generation (RAG): building AI agents that call tools, take independent actions, and verify their own outputs before they reach a user. Designs Role-Based Access Control (RBAC) and OAuth 2.0 / OpenID Connect (OIDC) authentication directly inside retrieval and agent execution paths, not bolted on afterward, and builds the faithfulness and verification layers that keep multi-agent and RAG systems auditable in production.

EXPERIENCE

AI & Automation Intern **Jun 2026 – Aug 2026**
White Cape Technologies *Sousse, Tunisia*

Role-Aware Agentic Retrieval-Augmented Generation (RAG) Knowledge Assistant

- Built an internal AI assistant so engineering teams can query company knowledge and codebases in plain language instead of manual search — architected as a 9-node LangGraph state machine (classify → retrieve → rerank → relevance gate → generate → faithfulness check) returning grounded, cited, role-aware answers streamed live over Server-Sent Events (SSE).
- Enforced role- and admin-scoped access inside the retrieval query itself rather than the UI, so unauthorized passages are never retrieved instead of being filtered out afterward — zero authorization violations across 280 audited retrievals, with restricted compensation content correctly withheld from non-admin accounts while remaining answerable for admins, behind Keycloak OpenID Connect (OIDC) session auth with Cross-Site Request Forgery (CSRF) protection.
- Built a labeled retrieval evaluation suite (held-out splits, reproducible offline scoring) and used it to set the relevance cutoff on measured evidence: 23 purpose-built near-miss cases ruled out a candidate threshold that accepted 12 of 29 unanswerable questions, settling the gate at 0.5 — where balanced accuracy peaks — and halving wrongly answered questions to 6 of 29.
- Diagnosed a gap in answer verification by probing the faithfulness layer with a fabricated claim, which scored a perfect 1.000 — identical to a grounded control — proving lexical-overlap checking cannot distinguish supported from invented statements, and delivered the probe as a reproducible regression test so any entailment-based replacement can be validated in seconds.
- Cut retrieved context per query by 46% by sweeping six candidate pruning thresholds and settling at 0.025, the value that maximized reduction while holding direct-evidence recall at 1.000.
- Removed a single point of failure in document ingestion — ~472 sequential model calls per document, where one failure discarded the entire run — by adding three-attempt exponential-backoff retry at the single LLM transport boundary, scoped to an allowlist that excludes Celery's soft-timeout signal so a timed-out job can still record its own failure.
- Ran a fully local inference stack (LiteLLM → Ollama, Qdrant, BGE-M3 embeddings, bge-reranker-v2-m3 via Hugging Face TEI), keeping all proprietary document content on-network for compliance.
- Developed the system itself through a three-role AI build loop (planner, generator, evaluator) across 45 contract-driven modules and 415 tests, verifying each fix with mutation probes so every guard is proven to fail when its defect is reintroduced.
  Technologies: LangGraph, FastAPI, LiteLLM, Ollama, Qdrant, BGE-M3, Keycloak, OIDC, Celery, Redis, pytest, Python, retrieval evaluation

AI & Automation Intern **Jun 2025 – Aug 2025**
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UX Insight Platform

- Built an AI-driven user-experience (UX) audit platform that automatically evaluates web and application interfaces from raw screenshots and interaction data, producing structured, evidence-backed UX reports for product teams without manual review.
- Built a computer-vision layer for UI element detection that classifies interface elements directly from screenshots, giving the platform machine-readable structure to reason over instead of raw pixels.
- Combined that detection output with Optical Character Recognition (OCR) text extraction and interaction telemetry to obtain structured, actionable UX findings for product teams, instead of screenshots they'd have to interpret manually.
- Grounded generated insights in a RAG layer backed by ChromaDB, retrieving relevant UX heuristics and prior findings before generation, to obtain professional, consultant-level reports rather than generic LLM output.
- Made the LLM layer provider-agnostic by routing across OpenAI, Mistral, DeepSeek, OpenRouter, and local Ollama models, so the platform keeps working if a single provider goes down or becomes too costly.
- Moved heavy AI work off the request path into an async job lifecycle with queueing, polling, and recovery, and secured the gateway with Keycloak OAuth 2.0 / OIDC and JSON Web Token (JWT) validation.
- Isolated LLM-inference, OCR, and image-processing workloads into fallback-aware FastAPI microservices behind the gateway, backed by PostgreSQL, Redis caching, and Docker Compose.
  Technologies: Angular, Spring Boot, FastAPI, ChromaDB, Keycloak, JWT, PostgreSQL, Redis, Docker Compose

PROJECTS

Agentic Video Production Pipeline **2026**

- Built an agentic video-production pipeline that runs unattended end-to-end, inspecting filesystem state before each step to skip already-completed work instead of reprocessing from scratch.
- Designed a self-correcting generation loop that detects rejected or malformed outputs and automatically reformulates the prompt and retries, rather than failing the run.
- Made the pipeline resumable after interruption, picking up mid-run instead of restarting.
- Hardened the browser-automation steps with Playwright selector fallbacks and retry logic, so the pipeline survives UI changes on the sites it automates.
  Technologies: Python, Playwright, FFmpeg, Whisper, voice-cloning text-to-speech (TTS)

MCP Multi-Agent Build Pipeline **2026**

- Built Model Context Protocol (MCP) servers exposing personal project tooling as callable tools inside Claude Code, letting agents call functions and take actions directly against real project files.
- Designed a multi-agent build pipeline with distinct planner, generator, and evaluator roles that write and check code against

contract tests rather than relying on manual diff review.
- Runs in active daily use as part of my own development workflow.

Technologies: MCP, Claude Code, Python

AI Image Vectorization Pipeline **2025**

- Designed a Python AI image-generation and vectorization pipeline for apparel design workflows under 8GB video RAM (VRAM) constraints.
- Automated preprocessing, Segment Anything Model (SAM) segmentation, Oklab color simplification, and Scalable Vector Graphics (SVG) tracing into repeatable vector-ready outputs.
- Isolated each pipeline stage in its own subprocess with explicit VRAM flushing between stages to obtain stable runs on consumer hardware instead of the fragmentation that comes from running everything in one long-lived process.
- Added batch checkpointing so a crash or interruption partway through a batch resumes from the last completed image, not from the start.

Technologies: Python, Segment Anything (SAM), VTracer, OpenCV, Pillow

QuickFlow — AI Productivity Assistant **2025**

- Built a full-stack AI productivity assistant by integrating Spring AI into a Spring Boot backend to obtain automated meeting summaries, structured minutes, and professional email drafts directly from raw meeting input.
- Owned the identity and security branch by configuring Keycloak OIDC with Google and Microsoft identity brokering alongside Time-based One-Time Password Multi-Factor Authentication (TOTP/MFA), obtaining secure, social-login-ready authentication shared across the whole team's workflows.
- Built the Angular frontend workflows by decoupling authentication state from in-progress work, so a session refresh never loses a user's unsaved draft.
- Designed reusable communication automation flows by templating the summary-to-email conversion step into a component the team reused across three features without rewriting it.

Technologies: Spring Boot, Spring AI, Angular, Keycloak, OIDC, JWT, TOTP, MFA

SKILLS

Agentic AI & LLM Orchestration: LangGraph, LangChain, Model Context Protocol (MCP), multi-agent orchestration, tool-calling & autonomous actions, Retrieval-Augmented Generation (RAG), Graph-RAG, LiteLLM, local LLM deployment (Ollama), reranking (TEI), semantic caching, faithfulness/verification checks, agent observability & audit logging, LLM inference optimization, prompt / context engineering
AI Security & Access Control: OAuth 2.0 / OpenID Connect (OIDC), Role-Based Access Control (RBAC), JWT validation, Keycloak, identity brokering (Google/Microsoft), multi-factor authentication (TOTP/MFA), AI/ML security (adversarial robustness, RAG poisoning defenses, vulnerability detection)
Backend & Data: FastAPI, Spring Boot, PostgreSQL, Redis, Qdrant, Vector Databases, ChromaDB, Celery, Server-Sent Events (SSE), microservices architecture
DevOps & Production Engineering: Docker & Docker Compose, Git/GitHub/GitLab, Linux CLI, contract-driven automated testing
Generative AI & Computer Vision: Diffusion Models (FLUX), Segment Anything (SAM), OCR, YOLO
Programming Languages: Python, Java, JavaScript/TypeScript, SQL, C, C++

CERTIFICATIONS

NVIDIA Deep Learning Institute (DLI): Building RAG Agents with LLMs · Building Transformer-Based NLP Applications · AI Cybersecurity Pipelines · Building LLM Applications With Prompt Engineering
Fundamentals of Deep Learning · Generative AI with Diffusion Models · Building Real-Time Video AI Applications · Applications of AI for Predictive Maintenance
DeepLearning.AI: Neural Networks and Deep Learning · Supervised Machine Learning
Google Cloud: Introduction to Generative AI
Competition: IndabaX Tunisia 2025 Challenge 1 — top 50% (31/69), 20-hour applied machine learning hackathon on 4G/5G telecom KPI anomaly detection

EDUCATION

École Nationale d'Ingénieurs de Sousse (ENISO) **Sep 2024 – Present**
Engineering Degree in Software Engineering

Preparatory Institute for Engineering Studies of Monastir **Sep 2022 – Jun 2024**
Mathematics & Physics Preparatory Cycle — National Rank: 509 / 1800

LANGUAGES

Arabic: Native **English:** Fluent **French:** Fluent **German: A1**